Emir Benarbia

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Data Engineering & AI student building RAG systems, LLM agents, and data pipelines

Seeking 6-Month Final-Year Internship (PFE) | **Jan – Jul 2027** | Open to Relocation

Education

National Engineering Degree – Data & Software Engineering **Sep. 2023 – Jun. 2027**

Faculty of Sciences of Bizerte · University of Carthage *Bizerte, Tunisia*

- Ranked **5th of 36** in the Data & Software Engineering program

Experience

Data & Software Engineering Intern **Jun. 2026 – Aug. 2026**

TcodeX *Tunis, Tunisia*

- Engineered **MediDocs**, a full-stack platform (Django, PostgreSQL, React) turning clinical scans into structured JSON.
- Benchmarked local and API-hosted **vision-language models** (incl. Nemotron-3-Nano-Omni-30B via OpenRouter) on Kaggle, then deployed the best-performing OCR pipeline to production.
- Built an **extraction-quality score** (TF-IDF/cosine similarity, edit distance) to quantify human correction effort.

Software Engineering Intern **Jun. 2025 – Aug. 2025**

Générale Tunisienne de l'Informatique (GTI) *Tunis, Tunisia*

- Built a credit platform (Angular, Spring Boot, PostgreSQL) for 3 user roles, **cutting processing time by 40%**.
- Built credit-lifecycle REST APIs with JUnit tests and an Angular frontend with RBAC and live status tracking.
- Deployed a local **Ollama** assistant that guides users through credit applications, reducing support-ticket volume.

Projects

Claim-Verify RAG | *Python, LangGraph, pgvector, FastAPI, React* **Sep. 2026 – Oct. 2026**

- RAG pipeline that fact-checks its own answer: splits it into atomic claims and verifies each one against every source.
- Per-claim verdicts: supported, contradicted, unverifiable or **contested** (sources disagree); **9/10** on a gold set.
- Hybrid search (pgvector + BM25) with a cross-encoder reranker: **evidence recall raised from 23/44 to 37/44**.

Mahara Match – *Technical Lead* | *FastAPI, PostgreSQL, OpenAPI, Docker, CI/CD* **Sep. 2026 – Present**

- Matches job seekers, employers and training providers through a shared ESCO-based taxonomy of **208 skills**.
- Designed a **contract-first microservice architecture**: 6 OpenAPI contracts, mock servers, one schema per service.
- Set up CI and a **golden-path end-to-end test** (CV upload → match → hire → dashboard update).

Fuel Station Monitoring System | *Python, Kafka, FastAPI, Prophet, LLM Agents, RAG* **Apr. 2026**

- Team of 3, extended solo: built the backend and Prophet forecasting, then **migrated ingestion to Kafka**.
- An autonomous **LLM agent** acts on critical alerts; a RAG assistant answers questions over past reports.

Global Education Intelligence Platform (BI) | *Python, PostgreSQL, Django, GitHub Actions* **Feb. 2026**

- Designed a **star-schema warehouse** fed by a dependency-aware ETL pipeline, rebuilt nightly on GitHub Actions.
- Served through a Django REST API with React/Streamlit dashboards, containerized with Docker.

Technical Skills

AI & Data: LLM Agents, RAG, Hybrid Search (BM25), Reranking, LangGraph, pgvector, PyTorch, Scikit-learn, NLP
Data Engineering: ETL Pipelines, Data Warehousing (star schema), Kafka, Spark, Hadoop, HBase, MLOps (MLflow, DVC)
Languages: Python (Pandas), SQL, Java, Go, JavaScript, C++
Backend & Web: Django, FastAPI, Spring Boot, React, Angular, REST APIs, Flask, Streamlit
Databases & DevOps: PostgreSQL, MongoDB, Docker, GitHub Actions (CI/CD), Git, Linux

Leadership

Vice President & Social Media Manager – Interact Club North Africa: led 15+ members.

Languages

Arabic: Native | **French**: Fluent (language of instruction) | **English**: Fluent (full professional proficiency)